

Steven Lio

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Summary

Machine Learning and AI specialist with 5+ years of experience building predictive models, automating data pipelines, and designing end-to-end ML workflows that improve decision-making and operational efficiency. Adept at translating complex findings into actionable business insights through effective visualization and communication.

Education

University of British Columbia

M.S. in Data Science

Vancouver, BC

Sep, 2021 - Aug, 2022

McMaster University

Honors B.Sc. in Mathematics and Statistics Co-op

Hamilton, ON

Sep, 2011 - Jun, 2016

Experience

Data Scientist

Sep, 2024 - Present

NielsenIQ

Toronto, ON

- Designed and deployed scalable end-to-end Machine Learning pipelines, leveraging Airflow and Snowflake for automated data ingestion, transformation and deployment workflows that accelerating client onboarding.
- Fine-tuned pre-trained embedding LLMs and applied NLP techniques to enable multi-language, cross-product recommendations for targeted marketing and personalized promotions across global CPG retailers.
- Improved receipt data extraction by incorporating LLM-based text processing with DSPy, correcting OCR errors to generate high-accuracy structured data for analytics and personalized marketing.
- Collaborated with client-facing execution teams to interpret complex ML results, understand client needs, and deliver customized, data-driven solutions that informed strategic actions.

Data Scientist

Apr, 2021 - Sep, 2024

Blueprint Software System

Toronto, ON

- Developed an intelligent recommendation system in Python to enhance customer's RPA processes efficiency.
- Optimized RPA conversion algorithms using Transformer-like ML techniques, achieving a 95% conversion rate.
- Developed tailored backend solutions using .NET framework and C# within an Agile environment.
- Performed A/B testing to optimize UI visualizations and improve user experience.
- Developed data integration processes and constructed probability models for data abnormally detections.

Data Scientist

Jan, 2020 - Dec, 2020

Bell Mobility

Mississauga, ON

- Predict customer lifetime value using multivariate generalized linear models and devised effective marketing campaign strategies to identify high-value market segments, enhancing customer acquisition quality.
- Applied NLP techniques such as LDA, classification models to support sentiment analysis on customer surveys.
- Modeled causal impact on time series data to evaluate marketing campaign effectiveness.
- Engineered a scalable MS SQL pipeline to process large calling detail records, generating new features and tripling customer attributes and strengthening downstream analytics and ML insights.

Market Insight Analyst

Aug, 2017 - Dec, 2019

Bell Mobility

Mississauga, ON

- Provided actionable insights to drive strategic decisions and business growth for marketing and product teams.
- Estimate market size and revenue utilizing sales and spatial data, and proposed optimal sales channel strategies.
- Analyzed promotional response, assessed product category elasticity to identify cross-selling opportunities.
- Developed multi-class classification models for churn prediction and enriched competitor analysis in Python.
- Supported Pre & Post campaign analyses, designing customized surveys to gain insights into customer behaviors.

Data QA Analyst

May, 2016 - Apr, 2017

Environment Canada

Burlington, ON

- Designed data QA process and digitized 40+ years of Canadian Great Lakes freshwater quality data.
- Identified anomalies and missing data patterns using statistical inference in R to maintain data integrity.
- Assisted project leaders to identify suitable benchmarks for developing data quality guidelines.

Data QA Support (Co-op)

May, 2015 - Dec, 2015

Environment Canada

Burlington, ON

- Developed automated data QA procedure in R and Excel VBA for data aggregation and outliers' detection.
- Generating dynamic summary reports on data quality and visualization on collected data trend.

Assistant Methodologist (Co-op)

Jan, 2014 - Aug, 2014

Statistics Canada

Ottawa, ON

- Assisted in researching and implementing a revised survey methodology in SAS for improving survey estimates.
- Managed the survey production process, including sample selection, regression analysis and estimations.

Skills

Programming Languages:  Python,  R,  C#,  SQL,  SAS,  HTML/  CSS, VB.Net,  Javascript

Machine Learning Libraries: scikit-learn, XGBoost, Spacy, PyMC, OpenCV, PaddleOCR, SciPy, Pandas, NumPy, NLTK

Deep Learning: PyTorch, Tensorflow/Keras, ONNX, Transformers

Agentic AI Development: Prompt-Engineering, Agent Evaluation, Google-ADK, DSPy, LangChain

Visualization: Dash, Shiny, matplotlib, Streamlit, ggplot, Plotly, Vega-Altair, Tableau, PowerBI

Tools and Frameworks: Snowflake, MS SQL Server, Docker, Airflow, Azure, PowerBI, PowerShell, Bash, ArcGIS, Git, Tableau, .Net, Jenkins, Microstrategy, $\text{\LaTeX}$